Project Report: Web Application Security Assessment

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1. Executive Summary

The objective of this project was to perform a Vulnerability Assessment and Penetration Testing (VAPT) on a target server. I successfully identified the target machine, discovered hidden directories, exploited a critical SQL Injection vulnerability, and monitored traffic using interception tools.

2. Technical Environment

- **Attacker Machine:** Kali Linux
- **Target Machine:** Metasploitable (IP: `10.0.2.4`)
- **Tools Used:** Nmap, Dirb, Burp Suite, Firefox

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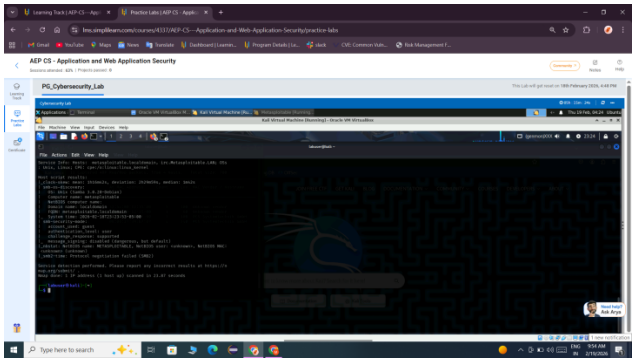
3. Phase 1: Network Scanning (Nmap)

Description: I used Nmap to discover the services running on the target IP. This helped identify that the machine was running a web server on port 80 and was identified as "Metasploitable."

- **Command:** `nmap -sC -sV 10.0.2.4`
- **Observation:** The scan confirmed the target OS as Unix and identified several open ports including FTP, SSH, and HTTP.



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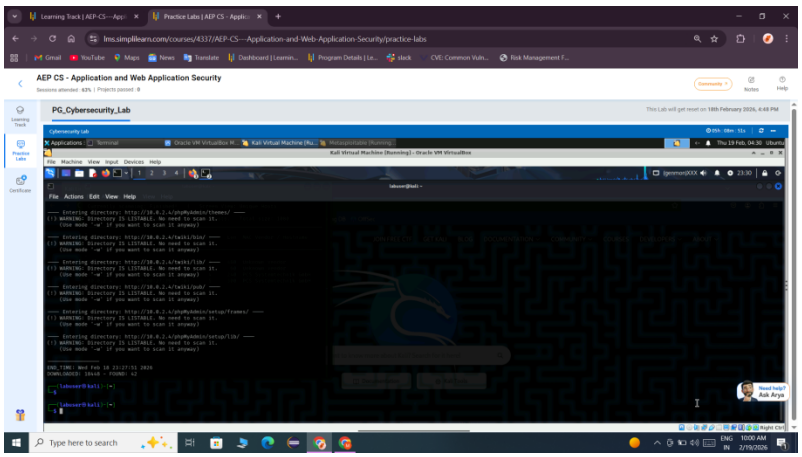
4. Phase 2: Directory Enumeration (Dirb)

Description: I used the Dirb tool to brute-force the web server for hidden files and directories that were not linked on the main page.

- **Command:** `dirb http://10.0.2.4`
- **Observation:** The tool found 42 items, including sensitive directories such as `/phpMyAdmin/` and `/dvwa/`.



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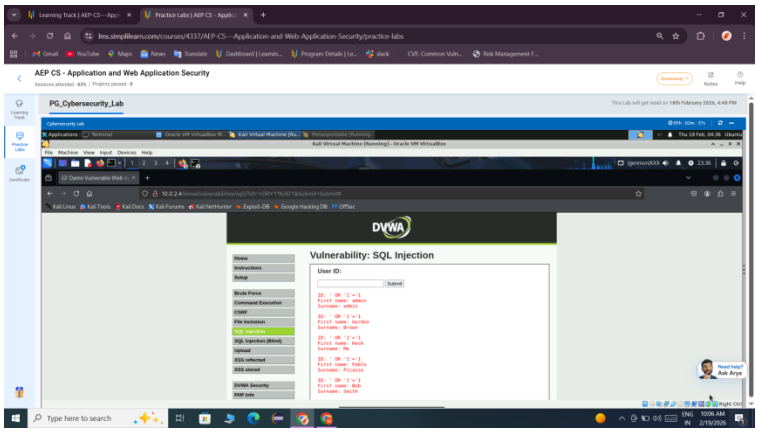
5. Phase 3: SQL Injection Vulnerability Exploitation

Description: I targeted the "Damn Vulnerable Web App" (DVWA) to test for SQL Injection. By entering a malicious payload into the User ID field, I bypassed the database logic.

- **Payload Used:** `' OR '1'='1`
- **Result:** The application revealed the First Name and Surname of every user in the database, proving a critical vulnerability exists.



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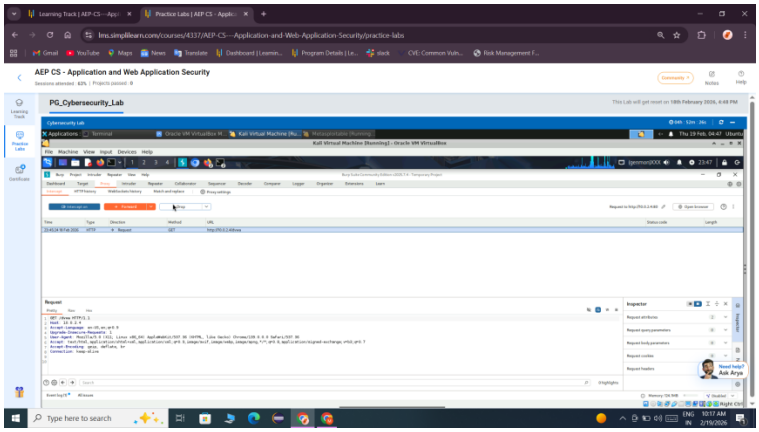
6. Phase 4: Traffic Interception (Burp Suite)

Description: I utilized Burp Suite to intercept an HTTP GET request between my browser and the target server to analyze the request headers.

- **Method:** Burp Proxy Intercept
- **Observation:** I successfully captured the raw HTTP request data, including the Host, User-Agent, and Cookie information.



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7. Conclusion

The assessment revealed that the target system is highly vulnerable to several web-based attacks. Specifically, the lack of input sanitization allowed for a successful SQL Injection, which could lead to a total data breach in a real-world scenario. It is recommended to implement prepared statements and restrict access to administrative directories.